A-Line Precision Alignment Tools

Price $10.00

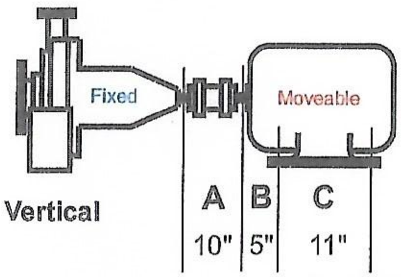
Back to Basics Instructional Graph Plotting Board

Learning How to Graph Plot

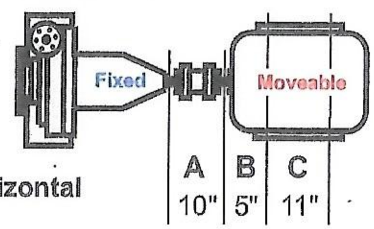
Graphical alignment is a technique that shows the relative position of the two shaft's centerlines on a piece of square grid graph paper.

First we must view the equipment to aligned in the same manner that appears on the graph plot. In this example we view the equipment with the "FIXED" on the left side and the "MOVEABLE" on the right. This will remain the same view both vertically and horizontally.

Each square ↔ =1.0", This measures the distance between the indicators and feet.



Vertical

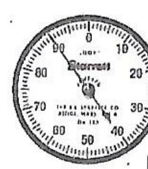


Horizontal

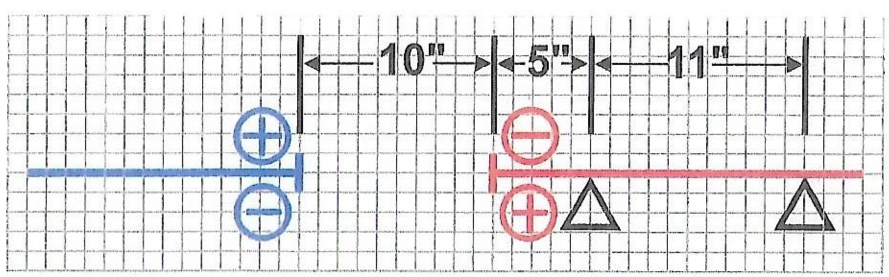
A. Distance between indicators
B. Distance to front foot
C. Distance to back foot

Each Square ↕ =.001", A dial indicator is used for this measurement

Plus (+) +10



Minus (-) -10



******To eliminate confusion the plus and**** minus signs should be marked on the graph**

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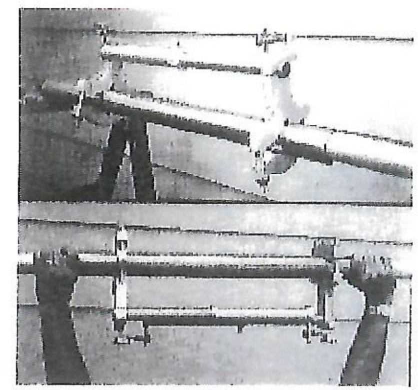
Preliminary

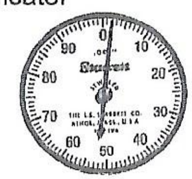
- ☐ Visual check of coupling, pipehangers, base bolts, coupling spacing etc.
- ☐ Check for coupling & shaft run out.
- ☐ Clean under feet, remove all rust, nicks, burrs and paint.
- ☐ Reduce to a minimum the number of shims.
- ☐ Check for soft foot.
- ☐ Check for sag.



Sag Check Example

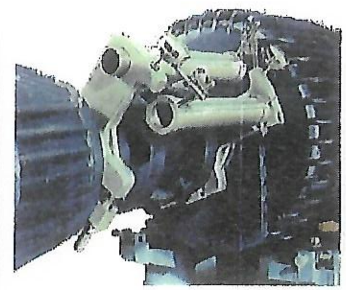
When allowing for Sag on vertical move, always start with a plus (+) reading.

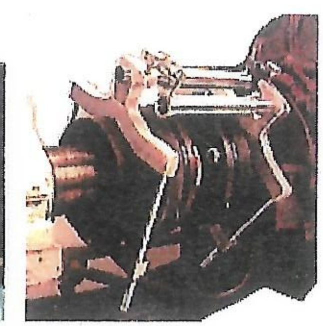
Example: .002" sag
Position the indicator to read +2.



Installation of Brackets

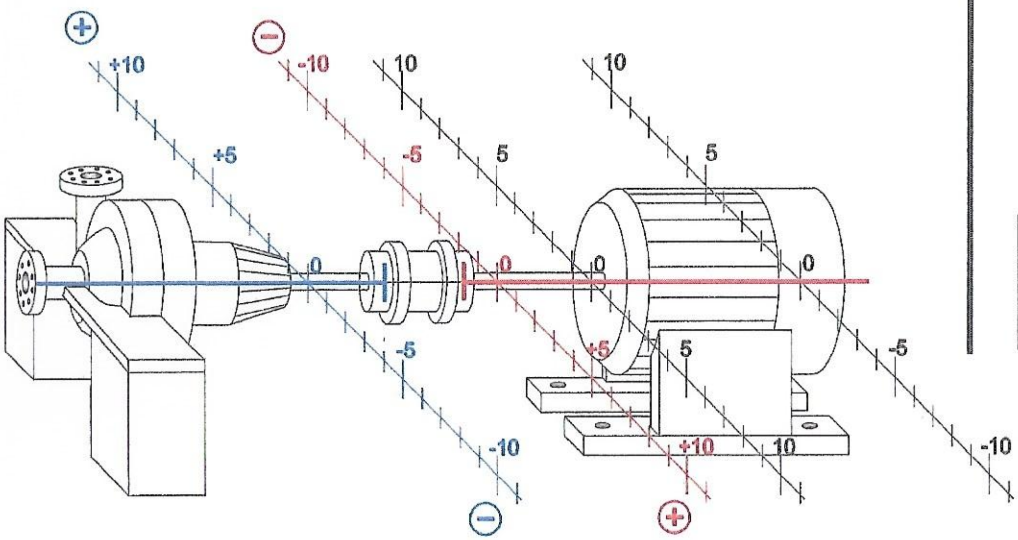
1. Clean mounting surface, file off all nicks and burrs.
2. Check indicators for sticking or loose needle.
3. Aim indicator stem directly toward center line of shaft.

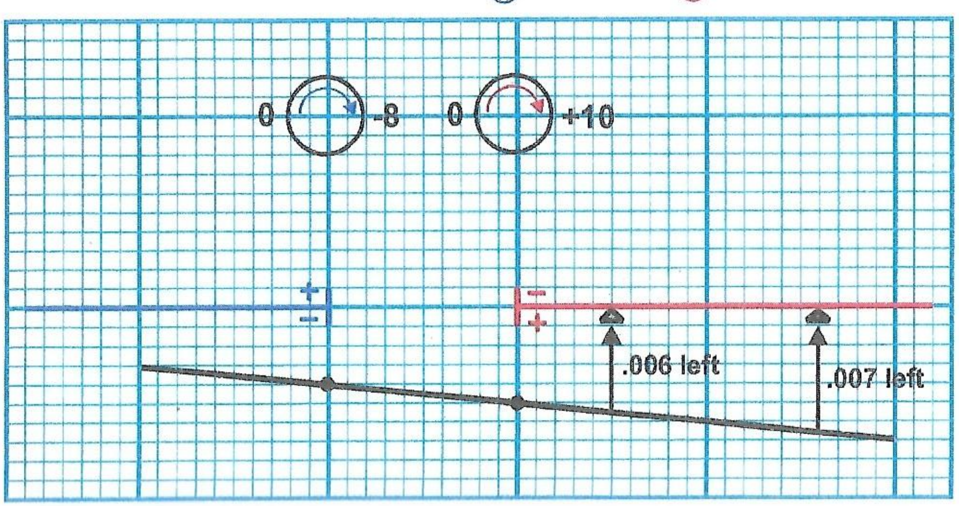




Horizontal Move

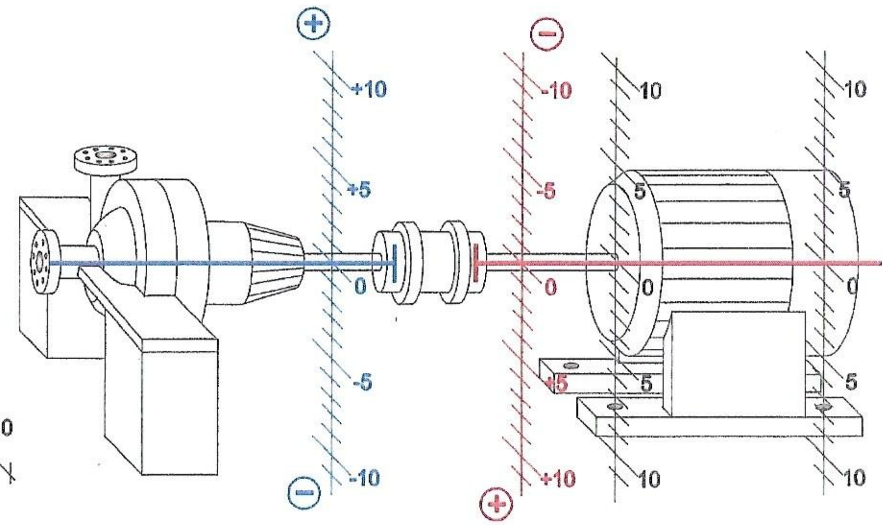
This example shows that if you viewed the machine from the pump end, zeroed the indicators on the left, rotate and read on the right. The indicator or the pump reads **-8** and the indicator on the motor reads **+10.** When graphed it shows that the motor shaft is **.004"** (1/2 T.I.R.) away from being collinear at the point where the indicator is mounted on the pump shaft and **.005"** out at the point where the indicator is on the motor shaft. When this line is projected back, it shows that the front feet of the motor need to be shifted **.006"** to the left, and the back feed need to be shifted **.007"** to the left.

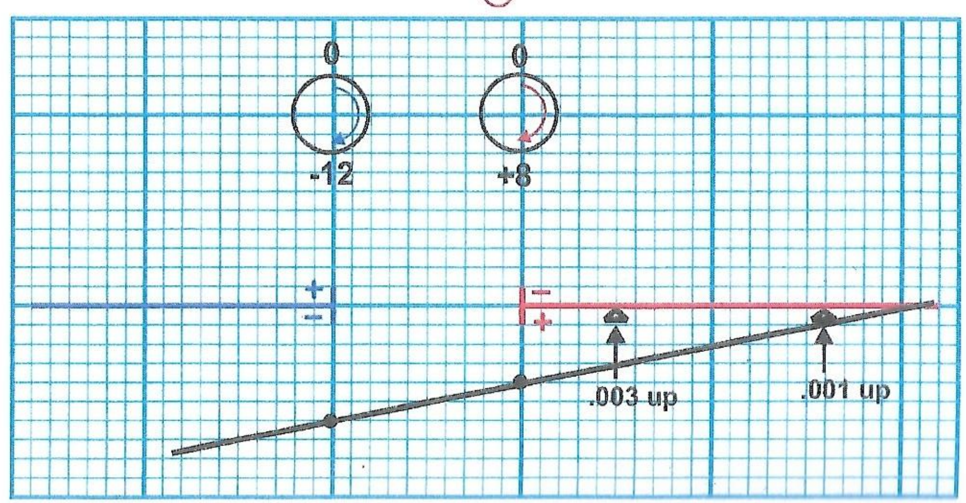




Vertical Move

The indicators are zeroed on the top and read at the bottom. In order to compensate for sag, start with a plus reading. The indicator on the pump reads **-12**, the indicator on the motor reads **+8**. This means that the shafts are one half the **T**otal **I**ndicator **R**eading (T.I.R.) from being collinear at these points. Using a piece of square grid graph paper to illustrate, these two points are marked and the line continued to the distance representing the feet of the motor. This shows that if the aligner adds **0.003"** shim to the front foot and a **0.001"** shim to the back foot of the motor, the motor shaft centerline will be collinear to the pump shaft centerline.

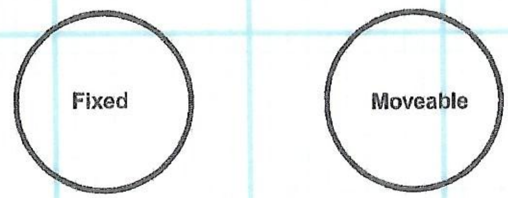




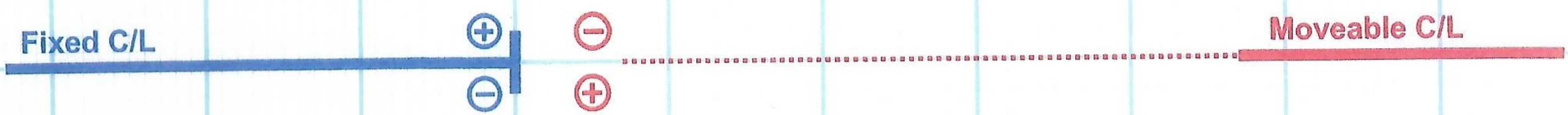
Alignment Report

Company:
Date:
Equipment:
Mechanic:

Distance A:
Distance B:
Distance C:



1. Extend Movable C/L to distance A.
2. Mark distance B & C at Movable C/L.
3. Mark 1/2 T.I.R. on appropriate
   side of centerline.
4. Extend line past foot location



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